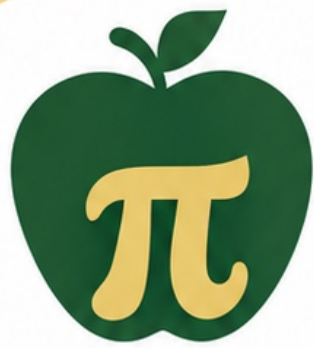




SCHOOL OF MATH

Think • Learn • Grow

Small Steps Big Futures

DECIMALS — TO THE TENTHS — AND HUNDREDTHS

A School of Math class on place value, reading decimals, comparing decimals, number lines, money, and measurement



Build confidence
with decimals



Connect math
to daily life



Develop precision
in money and
measurement



Clear explanations •
Strong foundations •
Engaging learning

Real Math. Brighter Futures.



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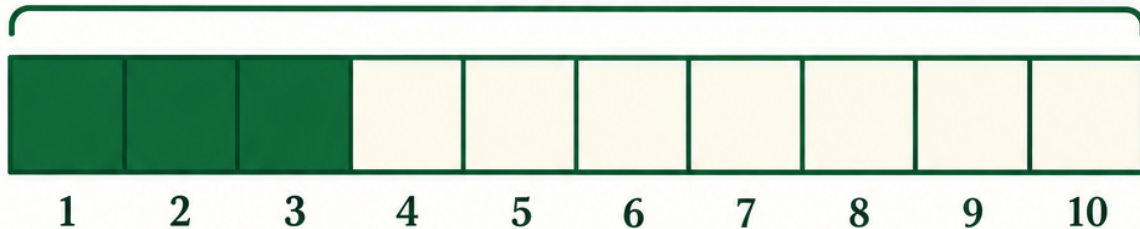


CLASS 13 • GRADES 3-4

Understanding Tenths

Divide one whole into 10 equal parts.

1 whole



3 of the 10 equal parts are shaded.

$$\frac{3}{10} = 0.3$$

three tenths

10 tells us how many equal parts make the whole.

3 tells us how many parts are shaded.

$$\text{One tenth} = \frac{1}{10} = 0.1$$

$$\text{Ten tenths} = 1 \text{ whole}$$

Decimals to the Tenths and Hundredths

Image 1 practice: Understanding tenths

Q1. Write $\frac{2}{10}$ as a decimal.

Q2. Write $\frac{6}{10}$ as a decimal.

Q3. Write 0.9 as a fraction with 10 on the bottom.

Q4. A strip has 10 equal parts. Seven are colored. What decimal of the strip is colored?

Q5. A strip has 10 equal parts. Two are colored. How many more parts should you color to show 0.4?

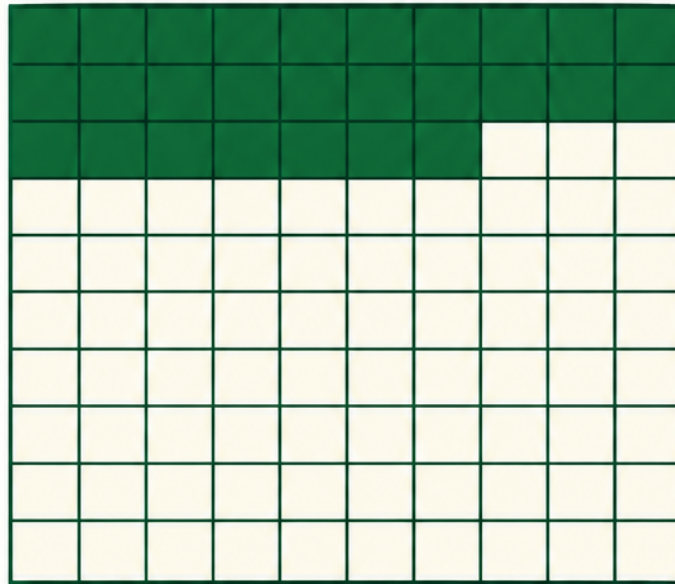


CLASS 13 • GRADES 3-4

Understanding Hundredths

Divide one whole into 100 equal parts.

1 whole = 100 equal squares



27 of the 100 squares are shaded.

$$\frac{27}{100} = 0.27$$

twenty-seven hundredths

100 tells us how many equal parts make the whole.

27 tells us how many parts are shaded.

One hundredth = $\frac{1}{100} = 0.01$ | 10 hundredths = 1 tenth

Decimals to the Tenths and Hundredths

Image 2 practice: Understanding hundredths

Q6. Write $12/100$ as a decimal.

Q7. Write 0.41 as a fraction with 100 on the bottom.

Q8. A grid has 100 equal squares. Eighteen are colored. What decimal of the grid is colored?

Q9. Write $4/100$ as a decimal.

Q10. A grid has 100 equal squares. Color 2 full rows of 10 and 3 extra squares. What decimal of the grid is colored?



CLASS 13 • GRADES 3-4

Reading & Writing Decimals

Understanding Place Value

2.45

Ones	Tenths	Hundredths
2	4	5

2 ones = 2

4 tenths = 0.4

5 hundredths = 0.05

STANDARD FORM**2.45****WORD FORM**two and forty-five
hundredths**EXPANDED FORM** $2.45 = 2 + 0.4 + 0.05$

The first place after the decimal point is tenths.
The second place is hundredths.

Watch the zero!

2.05 = two and five hundredths
The zero holds the tenths place.

Decimals to the Tenths and Hundredths

Image 3 practice: Reading and writing decimals

Q11. Write 0.2 in words.

Q12. Write 0.43 in words.

Q13. Write 'one and six tenths' as a decimal.

Q14. Write 'three and eight hundredths' as a decimal.

Q15. In 4.26, what is the value of the 2? What is the value of the 6?



CLASS 13 • GRADES 3-4

Equivalent Decimals

Different ways to write the same value.

$$0.5 = 0.50$$

five tenths fifty hundredths

1 whole = 10 equal parts



5 of 10 parts are shaded: $5/10 = 0.5$

Each tenth equals 10 hundredths.

5 tenths = 50 hundredths

$$\frac{5}{10} = \frac{50}{100}$$

A zero at the end of a decimal does not change its value.

MORE EXAMPLES

$$0.3 = 0.30 \quad | \quad 0.8 = 0.80 \quad | \quad 1.7 = 1.70$$



Position matters!

$$0.5 \neq 0.05$$

5 tenths is not 5 hundredths.

Decimals to the Tenths and Hundredths

Image 4 practice: Equivalent decimals

Q16. Complete the matching decimal: $0.2 = 0.\underline{\quad}$.

Q17. Write 0.60 using just one digit after the decimal point.

Q18. Choose the decimal equal to 0.9: 0.09 or 0.90.

Q19. Write 1.4 with two digits after the decimal point.

Q20. Are 0.7 and 0.07 equal? Explain using tenths and hundredths.



CLASS 13 • GRADES 3-4

Comparing and Ordering Decimals

Use place value to decide which is greater.



less than



greater than



equal to

Compare 0.35 and 0.5

- 1** Write the same number of decimal places:
 $0.5 = 0.50$

- 2** Compare tenths first:
3 tenths < 5 tenths.

$$0.35 < 0.50$$

So, $0.35 < 0.5$.

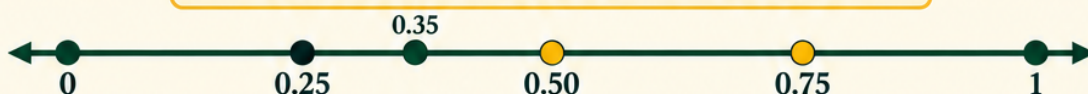
If tenths are equal, compare hundredths.

$$0.64 > 0.62$$

Same tenths; 4 hundredths > 2 hundredths.

Order from least to greatest.

$$0.35 < 0.50 < 0.75$$



Numbers increase to the right.



More decimal digits do not always mean a greater value.

Decimals to the Tenths and Hundredths

Image 5 practice: Comparing decimals

Q21. Which is greater: 0.2 or 0.6?

Q22. Fill the blank with $<$, $>$, or $=$: $0.31 \underline{\hspace{1cm}} 0.38$.

Q23. Fill the blank with $<$, $>$, or $=$: $0.4 \underline{\hspace{1cm}} 0.40$.

Q24. Which is greater: 0.7 or 0.58?

Q25. Put these decimals in order from least to greatest: 0.9, 0.24, 0.6.



CLASS 13 • GRADES 3-4

Decimals in Everyday Life

Use decimals for money and measurement.

MONEY



100 cents = 1 dollar

75 cents = \$0.75

Read: seventy-five cents.

LENGTH



100 centimeters = 1 meter

**1 meter + 25 centimeters
= 1.25 meters**

Read: one and twenty-five hundredths meters.

CAPACITY



1,000 milliliters = 1 liter

500 milliliters = 0.5 liter

Read: five tenths of a liter.



The unit tells us what the decimal measures.

Always include the unit: dollars, meters, or liters.

Decimals to the Tenths and Hundredths

Image 6 practice: Decimals in everyday life

Q26. A sticker costs 24 cents. Write its price in dollars.

Q27. A pencil costs \$0.63. How many cents is that?

Q28. A ribbon is 42 centimeters long. Use $100 \text{ centimeters} = 1 \text{ meter}$ to write its length in meters.

Q29. A bottle is labeled 0.6 liter. Read the amount in words.

Q30. One eraser costs \$0.4 and another costs \$0.40. Do they cost the same amount?

Decimals to the Tenths and Hundredths

Homework review: R1–R5

R1. Image 1: A strip has 10 equal parts and 4 are shaded. Write the shaded part as a decimal.

R2. Image 2: A grid has 100 equal squares and 16 are shaded. Write the shaded part as a decimal.

R3. Image 3: Write 'two and thirty-one hundredths' as a decimal.

R4. Image 4: Write a decimal equal to 0.6 with two digits after the point.

R5. Image 5: Which is greater: 0.8 or 0.73?

Decimals to the Tenths and Hundredths

Homework review: R6–R6

R6. Image 6: A bookmark costs 58 cents. Write the price in dollars.

Decimals to the Tenths and Hundredths

Homework word problems: W1–W5

W1. Lia colors 2 of the 10 equal parts of a flag. What decimal of the flag is colored?

W2. A class fills 9 of the 10 equal sections of a display. What decimal of the display is filled?

W3. A mosaic has 100 equal tiles. Thirty-two are yellow. What decimal of the mosaic is yellow?

W4. A design has 100 equal squares. Six are blue. What decimal of the design is blue?

W5. Nora finds a price tag that says 47 cents. How should she write that price in dollars?

Decimals to the Tenths and Hundredths

Homework word problems: W6–W10

W6. A toy car travels 1.23 meters. How would you read its distance in words?

W7. A jug label says 'two and four tenths liters.' What decimal belongs on the label?

W8. A ribbon is 68 centimeters long. If 100 centimeters make a meter, what is its length in meters?

W9. One bottle holds 0.2 liter. Another holds 0.20 liter. Do they hold the same amount?

W10. A green ribbon is 0.46 meter long. A blue ribbon is 0.49 meter long. Which ribbon is longer?

Decimals to the Tenths and Hundredths

Homework word problems: W11–W14

W11. One bookmark costs \$0.6. Another costs \$0.54. Which bookmark costs less?

W12. A red string is 0.7 meter, a yellow string is 0.32 meter, and a white string is 0.5 meter. Put them in order from shortest to longest.

W13. Jules writes \$0.08 for a sticker that costs 80 cents. Is the label correct? Write the correct price.

W14. Three bottles hold 0.4 liter, 0.40 liter, and 0.36 liter. Which two hold the same amount, and which holds the least?